

Business Requirements Document

Project name: HealthFirst Care – Patient Experience & Operational Efficiency Improvement Initiative

Project Overview

HealthFirst Care is undertaking an initiative to improve the patient experience and operational efficiency across its clinical and administrative departments. The project responds to recurring issues identified through stakeholder consultation and operational data review: long patient wait times, inefficient allocation of staff and equipment, and communication gaps across departments (scheduling, clinical, and IT).

This Business Requirements Document (BRD) consolidates input from patients, doctors, nurses, administrative staff, and IT teams, together with a review of appointment, feedback, and resource-utilisation data, to define the business requirements for a solution that streamlines scheduling, improves resource visibility, and strengthens communication between departments and with patients.

Background and Problem Statement

HealthFirst Care's stakeholders consistently report three interrelated problems, corroborated by the appointment, feedback, and resource datasets reviewed for this document:

- **Long patient wait times:** Patients report waits of 30–60 minutes past scheduled times and a lack of updates during delays. This is consistent with the appointment log, where 12 of 20 sampled appointments (60%) ended in a non-completed outcome (No Show, Rescheduled, or Cancelled), indicating scheduling friction that compounds delays for remaining patients.
- **Appointment scheduling data (n = 20 records):** Only 8 of 20 appointments (40%) were marked Completed; the remaining 60% were split evenly across No Show, Rescheduled, and Cancelled (4 each, 20% apiece). Each department in the sample runs a single fixed time slot (e.g., Cardiology at 10:00 AM, Orthopedics at 11:00 AM), which combined with a high non-completion rate points to scheduling rigidity and conflicts consistent with the double-booking issues Maria Carter (Appointment Scheduler) described.

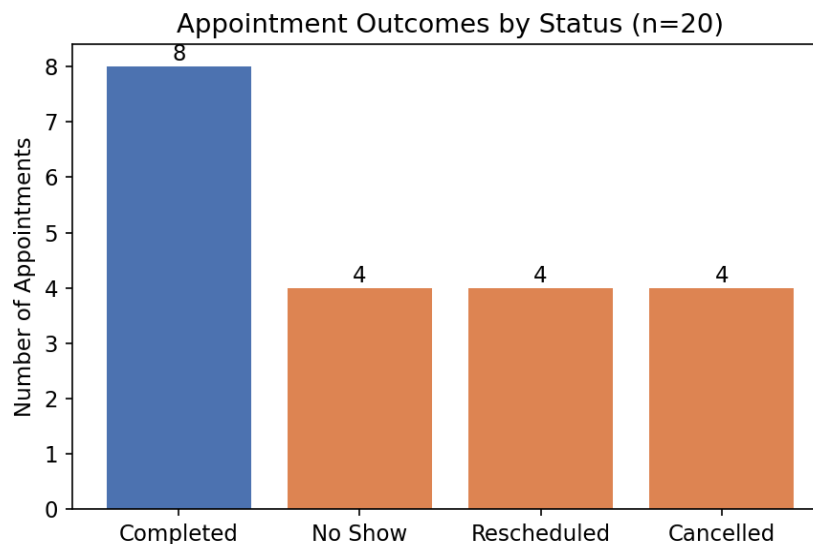


Figure 1 — Appointment outcomes by status. Source: *appointment_data.csv* (n=20).

- **Patient feedback data (n = 15 records):** The overall average satisfaction score is 7.0 out of 10. Cardiology and Neurology score highest (8.33 average each), while General Medicine (5.33 average) and Paediatrics (6.0 average) score lowest. Six of fifteen responses (40%) scored 6 or below, with comments including "Long wait times," "Delayed response," and "Room cleanliness issue" — directly corroborating the wait-time and communication concerns raised in the stakeholder profiles.

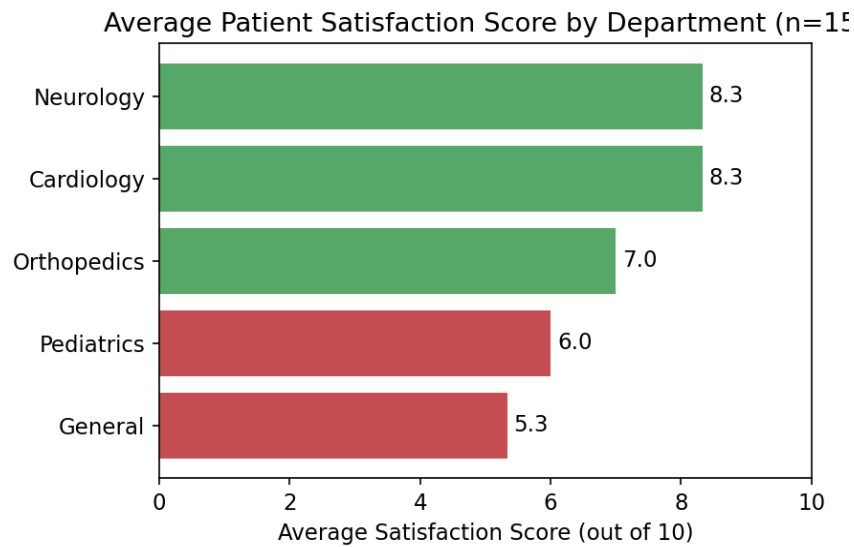


Figure 2 — Average satisfaction score by department. Source: *feedback_data.csv* (n=15).

- Resource utilisation data (n = 10 records):** Of the ten logged resource records, 4 show status "Available" (Doctor and Technician entries, 8 and 10 usage hours respectively), 2 show "Unavailable" (Nurse entries, 0 usage hours), 2 show "In Use" (Equipment, 6 usage hours), and 2 show "Under Maintenance" (Room, 2 usage hours). The recurring "Unavailable" nurse entries and "Under Maintenance" room entries in this sample are consistent with the staffing shortages and equipment-access constraints described by nursing and physician stakeholders.

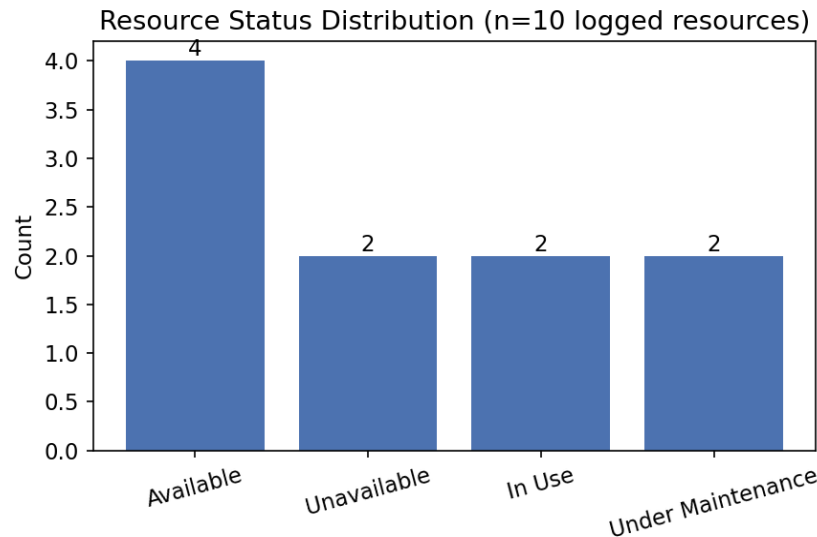


Figure 3 — Resource status distribution. Source: *resource_data.csv* (n=10).

- Inefficient resource allocation:** Doctors and nurses describe overbooked schedules, limited access to diagnostic equipment during peak hours, uneven nurse-to-patient ratios, and supply shortages. The resource dataset shows nursing resources logged as "Unavailable" with 0 usage hours on the sampled dates, and equipment/rooms shown "In Use" or "Under Maintenance" for a combined 40% of the ten logged resource records, limiting throughput during peak periods.
- Communication gaps across departments:** Doctors report delayed lab results and inefficient handoffs; nurses report poor coordination with the emergency department and delayed radiology results; administrative staff report

double bookings from limited visibility into doctor availability; IT staff confirm that scheduling and record-management systems are not integrated, creating data silos.

These issues have a measurable impact on patient satisfaction: average satisfaction scores in the reviewed feedback data fall to 5.3/10 in General Medicine and 6.0/10 in Paediatrics, both below the 7.0/10 overall average, with comments citing "long wait times," "delayed response," and "room cleanliness issue."

Project scope

In-scope

- Automated online appointment scheduling with real-time doctor/resource availability, to reduce double bookings and scheduling conflicts.
- Automated notifications (email/SMS) for appointment confirmations, cancellations, and delays.
- Integration between the scheduling system and existing patient record systems to close data silos flagged by IT.
- Dashboards/reporting for resource utilisation (staff, equipment, rooms) to support allocation decisions.
- Improved inter-departmental handoff and referral workflows (e.g., lab-to-clinician, physiotherapy-to-orthopaedics).
- Basic analytics on patient feedback and wait times to track the 20% wait-time reduction target.

Out-of-scope

- Replacement of the core Hospital Information System (HIS) or billing platform (only integration, not replacement, is in scope).
- Physical expansion of facilities (e.g., new operating rooms, additional beds).
- Hiring or staffing-level changes to nursing/medical rosters (may be recommended, but is a separate HR initiative).
- Telemedicine or virtual-consultation capability.
- Non-HealthFirst Care partner clinics or third-party referral networks.

Stakeholders

Stakeholder needs below are drawn directly from the Stakeholder Profiles document provided for requirement gathering.

Stakeholder Group	Representative Profiles	Primary Interest / Concern	Requirement Category
Patients	Patient #1; Patient #2	Easy scheduling, timely updates, shorter waits, clear post-visit communication	Functional (scheduling, notifications)
Doctors	Cardiology; Orthopaedics	Manageable schedules, timely diagnostic/lab results, smoother handoffs between departments	Functional (resource visibility, workflow)
Nurses	Paediatrics; Emergency	Adequate staffing and supplies, advance notice of transfers, better inter-department coordination	Functional (staffing/resource alerts)
Administrative Staff	Scheduler; Billing	Elimination of double bookings, real-time doctor availability, modern and accurate records	Functional (scheduling engine, records)
IT Teams	IT Manager; Developer	Integrated systems, reduced downtime, secure and user-friendly interfaces	Non-functional (integration, security, uptime)

Project Sponsor / HealthFirst Care Leadership	Project manager	Reduced wait times, improved patient satisfaction, controlled costs within budget	Business objectives / Constraints
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Business objectives
- Reduce average patient wait times by 20% within 12 months of go-live, addressing the wait-time complaints raised across patient profiles and reflected in appointment-completion patterns. - Reduce double-booking and scheduling-conflict incidents reported by administrative staff by providing real-time visibility into doctor and resource availability. - Improve inter-departmental communication and handoff efficiency (e.g., lab-to-clinician, radiology-to-ER, physiotherapy-to-orthopaedics) to reduce care delays. - Improve overall patient satisfaction scores, with particular focus on the lowest-performing departments identified in the feedback data (General Medicine and Paediatrics). - Consolidate scheduling and record-management data to eliminate the data silos identified by the IT team, within the constraints of the approved budget.

Requirements			
Functional requirements of this project.			
ID	Requirement	Description	Source Stakeholder(s)
FR-1	Scheduling	Provide an online appointment booking system with real-time visibility of doctor and resource availability to prevent double bookings and reduce scheduling conflicts.	Administrative staff (Maria Carter); Patients
FR-2	Notifications	Automatically notify patients via email/SMS of appointment confirmations, changes, and delays, including estimated wait-time updates.	Patients; IT (Laura Simkow)
FR-3	Resource visibility	Provide staff with a real-time view of equipment, room, and staff availability to support scheduling and reduce peak-hour bottlenecks.	Doctors; Nurses
FR-4	Handoff workflow	Support structured inter-departmental handoff and referral workflows (e.g., lab results, physiotherapy updates, ICU transfers) with status tracking.	Doctors; Nurses
FR-5	Records integration	Integrate the scheduling module with the existing patient record system to eliminate duplicate data entry and data silos.	IT Team (Rajesh Singh)
FR-6	Reporting	Provide reporting on wait times, resource utilisation, and patient satisfaction to support ongoing monitoring against business objectives.	Project Sponsor; Administrative Staff

Non-functional requirements			
ID	Requirement	Description	Source Stakeholder(s)
NFR-1	Usability	The scheduling interface must be simple enough for patients who find current systems confusing (e.g., Lak Ayer) while remaining efficient for staff.	Patients; Administrative Staff

NFR-2	Reliability/Uptime	The system must minimise network downtime affecting online scheduling and communication services.	IT Team (Rajesh Singh)
NFR-3	Security	The system must remediate known security vulnerabilities in legacy systems and protect patient data in line with applicable healthcare data-protection requirements.	IT Team (Laura Simkow)
NFR-4	Performance	Real-time availability and notification features must update without noticeable delay during peak booking periods.	Administrative Staff; Patients
NFR-5	Scalability	The solution should be cloud-capable to allow scaling as patient volumes grow, per IT recommendations.	IT Team (Rajesh Singh)

Assumptions

- Stakeholder profiles and appointment/feedback/resource datasets provided for this exercise are representative of recurring, real operational patterns at HealthFirst Care, even though the datasets themselves are small, simulated samples (20 appointment records, 15 feedback records, 10 resource records) rather than a full operational extract.
- Existing IT infrastructure can support integration with a new or upgraded scheduling module without a full core-system replacement.
- Clinical and administrative staff will be available for user-acceptance testing and training ahead of go-live.
- Patients have sufficient access to email or SMS to receive automated notifications; alternative channels may be needed for patients without reliable access.

Constraints

- Budget limitations for system upgrades restrict the scope to integration with, rather than replacement of, existing core systems.
- Any new system must work within existing hospital network infrastructure, which IT reports experiences frequent downtime.
- Implementation must avoid disrupting ongoing patient care, requiring phased rollout rather than a single "big-bang" cutover.
- Timeline and staff availability for training may be limited given existing staffing pressures reported by nursing staff.

Conclusion

The stakeholder profiles and supporting data reviewed for this document point to a consistent set of problems at HealthFirst Care: scheduling inefficiency and double bookings, uneven allocation of staff and equipment, and fragmented communication both with patients and across departments. The requirements defined in this document — an integrated, real-time scheduling and notification system, improved resource visibility, structured hand-off workflows, and consolidated records — are intended to directly address these issues within the stated budget constraint.

Successful delivery against the business objectives, particularly the targeted 20% reduction in average patient wait times, should be tracked using the same categories of data reviewed here (appointment status, patient satisfaction, and resource utilisation) so that progress can be verified against a consistent baseline.